Name & Surname **Polimi ID**

Exercise 1

A. Could you please summarize why manual parallelization is preferred to automatic parallelization? (1)

B. Please describe the differences between flow, output, and anti-dependencies. (1)

C. Consider the following nested loop:

```
for (i=0; i<100; i++) {
  for (j=0; j<100; j++)
    b[i][j] += f(a[i][j]);
  for (j=99; j>=0; j++)
    a[i][j] += f(b[i][j]);
}
```

Which loop can be parallelized and under which conditions (i and j are declared as integers)? Is there a way to improve the code? Is the OpenMP nowait clause a possible option? Justify your answers. (2)

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Exercise 2

A. Please describe the parallel gather pattern, including assumptions, PRAM-based complexity, and examples. (2)

B. Please discuss how the Split parallel pattern works. How is it implemented (pseudo code required and PRAM-based complexity)? (2)

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Exercise 3

A. What is the difference between a static and a dynamic schedule in the OpenMP for construct? (1)

B. What happens when a variable is declared as firstprivate in an OpenMP pragma? (1) **This question can be skipped if you got 1 or 3 points in the second challenge.**

C. Describe in detail the effects of the following OpenMP pragma. (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**

```
#pragma omp parallel for reduction(+:sum)
```

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Exercise 4

A. **Why** does MPI enable the usage of distributed architectures? (1)

B. In which **scenario** does it make sense to use MPI_ANY_TAG in `MPI_Probe`? (1)

C. What is the **purpose** of the function `MPI_Reduce`? When the input buffer in each MPI process has more than one value, what is the **size** and **content** of the output buffer? (2)